

JVM, JRE, JDK (Basics Remember Forever)

- **JDK** → Development kit (compiler + JRE).
- **JRE** → Environment to run Java code.
- **JVM** → Executes bytecode, platform-independent.
- **JVM** = classloader + memory + execution engine + GC.

One-line memory:

JDK > JRE > JVM (JDK includes everything).

Java Memory Areas

- **Heap** → Objects, instance variables.
- **Stack** → Method calls, local variables.
- **Method Area** → Class metadata.
- **PC Register + Native Stack** → Internal JVM work.

OOP in One Shot

- **Encapsulation** → Hide data using private + getters/setters.
- **Inheritance** → Code reuse using extends.
- **Polymorphism** → Many forms (method overloading + overriding).
- **Abstraction** → Hide implementation (abstract class, interface).

Important Keywords

- **static** → Shared across all objects.
- **final** → Variable constant, method cannot override, class cannot extend.
- **this** → Current object.
- **super** → Parent object.

Constructors Revision

- No return type.

- Runs automatically when object created.
- Default constructor added by compiler if none present.
- Can overload constructors.

Immutability (Quick Formula)

To make a class immutable:

1. Make class final.
2. Make fields private and final.
3. No setters.
4. Return copies of mutable fields.

String Concepts (Very Important)

- Strings are **immutable**.
- "abc" goes to **String Pool**.
- new String("abc") goes to **heap**.
- StringBuilder → Not thread-safe, faster.
- StringBuffer → Thread-safe, slower.

Object Class — Must Know Methods

- toString()
- equals()
- hashCode()
- clone()
- finalize()
- wait(), notify(), notifyAll()

Collections Revision

Must-Know Classes:

- **List** → ArrayList, LinkedList
- **Set** → HashSet, LinkedHashSet, TreeSet
- **Map** → HashMap, LinkedHashMap, TreeMap
- **Queue** → PriorityQueue

Must-Know Concepts:

- **HashMap internal** → array + linked list + tree
- **equals & hashCode** must be overridden together
- **ConcurrentHashMap** → safe for concurrency
- **Fail-fast** → ConcurrentModificationException
- **Fail-safe** → Works on copy (Iterator on concurrent maps)

Multithreading Essentials

- **Thread creation** → extend Thread / implement Runnable.
- **synchronized** keyword → lock.
- **volatile** → visibility guarantee.
- **ThreadLocal** → separate copy per thread.
- **ExecutorService** → Thread pool.

Exceptions (Ultra-Short)

- **Checked** → must handle (IOException).
- **Unchecked** → runtime (NullPointerException).
- try-catch-finally
- throw vs throws
- Custom exception = extends Exception/RuntimeException.

Functional Programming (Java 8 Quick Notes)

- **Lambda** → (a, b) -> a + b.

- **Method reference** → `System.out::println`.
- **Stream API** → `filter`, `map`, `reduce`, `collect`.
- **Optional** → Avoid null checks.

Important Java 8 Stream Methods

- `filter()`
- `map()`
- `flatMap()`
- `sorted()`
- `distinct()`
- `reduce()`
- `collect()`
- `findFirst()`
- `findAny()`
- `anyMatch()`, `allMatch()`, `noneMatch()`

Serialization (One Line)

- Converts object into byte stream.
- Mark with implements `Serializable`.

Design Patterns (Only Essentials)

- **Singleton**
- **Factory**
- **Builder**
- **Strategy**
- **Observer**

Concurrency Locks

- **ReentrantLock** → explicit lock/unlock.
- **Semaphore** → limit access.
- **CountDownLatch** → wait until tasks finish.
- **CyclicBarrier** → wait for each other.

JDBC Essentials

- Load driver → get connection → prepare statement → execute → close resources.

Java 8 Date/Time

- `LocalDate`
- `LocalTime`
- `LocalDateTime`
- `ZoneId`
- `DateTimeFormatter`

Garbage Collection Quick Notes

- Removes unused objects automatically.
- **Minor GC** → young gen.
- **Major GC** → old gen.
- Cannot force GC, but `System.gc()` suggests it.

Most Important One-Liners (Interview Gold)

- Java is **pass-by-value**, always.
- **Method overloading** → compile-time.
- **Method overriding** → runtime.
- `HashMap` allows **one null key**, multiple null values.

- Interface methods are **public & abstract** by default.
- Functional interface → exactly **one abstract method**.

JAVA Interview Buddy